Temporal and Spatial Changes in Apparent Fishing Activity within Marine Protected Areas, 2022–2024

An Analysis of CHN and UNKNOWN-CHN Categories Using Global Fishing Watch AIS Data

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Abstract

This study descriptively examines changes in Global Fishing Watch (GFW) apparent fishing activity within marine protected areas (MPAs) from 2022 to 2024, focusing on the CHN and UNKNOWN-CHN flag categories. Thirty-six monthly gridded data files covering January 2022 through December 2024 were processed, comprising 41.862.480 records. After overlapping protected-area geometries were combined and shared grid observations deduplicated, the analysis covered 150 analytical units. Apparent fishing hours for the CHN category increased from 142.376.55 hours in 2022 to 151.150.97 hours in 2024 (6.16%). UNKNOWN-CHN increased from 4.631.42 to 19.434.50 hours (319.62%). Across all 150 units, total fishing hours rose by 4.230.13 hours (2.54%) from 2023 to 2024, while AIS presence hours fell by 6.095.63 hours (1.55%). UNKNOWN-CHN accounted for approximately 83.7% of the net increase in apparent fishing hours. Changes were spatially concentrated: the three largest increases among units with positive fishing hours in 2024 represented 71.74% of all positive changes, while Marae Moana alone accounted for 69.49% of the absolute negative changes. Changes in fishing and AIS hours were strongly correlated at the unit level (Pearson r = 0.8501; n = 49), although their aggregate totals moved in opposite directions. Seasonal distributions also differed: CHN activity remained concentrated in the third quarter, whereas UNKNOWN-CHN activity clustered in the third and fourth quarters of 2024. These findings are AIS-based model estimates; they do not establish catch volume, vessel nationality, illegal fishing, or violations of MPA rules. Results should be interpreted in light of changes in AIS reception, uncertainty in flag assignment, grid resolution, data-processing versions, and MPA-specific regulations.

Keywords: marine protected areas; apparent fishing activity; Global Fishing Watch; Automatic Identification System; AIS; spatial overlap; fishing hours; UNKNOWN-CHN.

1. Introduction

Marine protected areas are established to conserve marine biodiversity, safeguard sensitive habitats, and manage human activities. However, designation of an area for protection does not by itself show that anthropogenic pressures have ceased or that conservation objectives have been achieved. Monitoring

the spatial and temporal distribution of vessel activity can help describe patterns of marine use and identify locations that warrant closer examination.

The Automatic Identification System (AIS) transmits vessel identity and position information for maritime safety and collision avoidance. AIS-based fishing-monitoring systems use vessel characteristics, movement patterns, and classification models to estimate periods during which a vessel may be apparently engaged in fishing. Global Fishing Watch (GFW) publishes apparent fishing activity data derived from AIS-based models and, where available, data provided by partners. These estimates are valuable for large-scale monitoring, but they are not equivalent to verified fishing events, catch volumes, fishing mortality, or enforcement findings (Kroodsma et al., 2018; Global Fishing Watch, n.d.-a).

AIS coverage is neither complete nor spatially uniform. Some vessels may not carry AIS, may not transmit continuously, or may have incomplete identity information. Signal reception can vary by location and over time. GFW has reported improved reception in some high-density areas of China and Southeast Asia following the introduction of dynamic AIS receivers around 2022. In addition, version 3 of the data-processing pipeline introduced classification changes that affect estimates of apparent fishing hours relative to the previous pipeline. Observed differences over time may therefore reflect factors beyond changes in fishing behavior.

This study descriptively examines apparent fishing activity within the MPA analytical areas over 2022–2024. Focusing on the CHN and UNKNOWN-CHN categories, it addresses four questions: (i) how did annual and seasonal apparent fishing hours change; (ii) how concentrated were changes across MPA analytical units; (iii) how did fishing hours change relative to AIS presence hours; and (iv) how might overlapping MPA geometries affect aggregate totals?

The use of AIS data in fisheries research has made it possible to monitor vessel positions regularly and at broad scales. An influential global study showed that fishing patterns derived from AIS signals can be examined at both temporal and spatial scales (Kroodsma et al., 2018). The more recent GFW data product estimates “apparent” fishing hours using classification models based on vessel identity and movement patterns; this measure is not a direct catch record or independent observation (Global Fishing Watch, 2025a). This distinction requires separating the detection of activity within protected areas from assessments of conservation performance or legal compliance.

The research gap is not limited to determining whether activity occurs within a protected-area boundary. Repeated counting of shared grid observations where protected-area records overlap, uncertainty in flag assignment, and changes in AIS reception between years can complicate comparative interpretation. This study addresses these issues by analyzing monthly data from 2022–2024 through analytical units that deduplicate shared cells, and by considering overall trends alongside examples of local concentration.

The study makes three contributions: (i) it compares annual and seasonal patterns for the CHN and UNKNOWN-CHN categories within a common framework; (ii) it reduces duplicate counting caused by intersecting MPA polygons by constructing analytical units; and (iii) it reports aggregate hours, AIS presence hours, and unit-level changes together, making the limits of interpretation explicit. The analysis is exploratory and descriptive; it is not designed to estimate causal effects or produce MPA performance scores.

2. Materials and methods

2.1. Data source and variables

The analysis used 36 monthly GFW fleet files covering January 2022 through December 2024. The filename pattern (fleet-monthly-csvs-10-v3-YYYY-MM-01.csv) is consistent with GFW's monthly static apparent fishing activity product at 0.1-degree resolution. Fields used included date, year, month, the latitude and longitude of the grid cell's lower-left corner, flag category, gear type, total AIS presence hours (hours), and model-estimated apparent fishing hours (fishing_hours). The analysis was restricted to CHN and UNKNOWN-CHN. UNKNOWN-CHN is a GFW category based on invalid or unresolved vessel identity and the fact that most apparent fishing activity is associated with China's exclusive economic zone; it is not evidence that vessels are Chinese-flagged or Chinese-owned.

2.2. Data validation and aggregation

The 36 monthly files contained 41.862.480 rows in total. Automated structural checks assessed date and numeric-field validity, consistency between the month in each filename and the record date, coordinate ranges, negative hour values, whether fishing_hours exceeded hours, and missing flag or gear fields. No records were flagged by these checks. This result indicates only internal file consistency; it does not establish complete AIS reception, accurate vessel identities, or the performance of the fishing-detection model.

The MPA cell mask contained 88.425 unique records by year–flag–grid key, with no duplicate keys. Annual totals derived from the monthly files were compared against independently deduplicated annual MPA totals. The largest absolute difference was 2.91×10^{-11} hours for fishing time and 5.82×10^{-11} hours for AIS presence time. These results indicate numerical agreement within floating-point precision.

2.3. MPA geometries and overlap correction

MPA geometries were evaluated for spatial intersection. The 62 site_pid records formed 23 connected overlap components (21 multi-record components and two single-record components), while 127 records were independent. The analysis therefore used 150 units. Within each component, shared grid observations were deduplicated using a composite key consisting of analytical unit, year, flag, cell latitude, and cell longitude. This prevents the same grid observation from being counted more than once because two or more MPA records overlap.

Overlap components are connected components: membership can be transitive, and not every pair of polygons in a component necessarily intersects directly. Two nearly identical MPA records for the Phoenix Islands Protected Area had a Jaccard overlap of approximately 97.27% and were treated as one analytical unit during aggregation. Unit-level fishing-hour totals matched global unique-cell totals. A 1.1505-hour difference in CHN AIS presence hours in 2022 arose from a shared boundary cell between two separate Norfolk singleton units; fishing hours in that cell were zero. This small difference was documented as a boundary case and was not interpreted as additional fishing activity.

2.4. Temporal and spatial summaries

Annual and monthly totals were calculated by flag category. Quarterly values were summed from monthly records, and each quarter's share was expressed as a percentage of that category's annual total. MPA-level change was calculated as the 2024 value minus the 2023 value. Two analytical populations were distinguished: all 150 units for overall net change, and the 49 units with positive fishing hours in 2024 for describing the distribution of positive and negative changes. These populations are not interchangeable. The 101 units with zero fishing hours in 2024 had recorded 655.56 hours in 2023, explaining the

difference between the +4.885.69-hour net increase in the 2024-positive subset and the +4.230.13-hour increase across all units.

The Pearson correlation between unit-level changes in fishing hours and AIS presence hours was calculated for the 49 units with positive fishing hours in 2024. This analysis describes an association and does not establish causality. The fishing/AIS ratio was calculated as apparent fishing hours divided by AIS presence hours and expressed as a percentage. This ratio is not a catch-per-unit-effort or compliance measure. Cells with zero fishing hours in 2023 and positive fishing hours in 2024 were defined as newly observed positive activity at the grid-cell scale; this does not show that fishing began in the real world for the first time in 2024.

Analytical units were constructed by combining each connected component of MPA records in the geometric-intersection graph with the non-intersecting singleton records. Thus, 62 site_pid records were represented in 23 components and 127 records as independent units, producing 150 analytical units in total. This grouping does not imply statistical independence and does not turn each component into a single administrative management area. Unit-level results are therefore treated primarily as descriptive summaries that reduce duplication caused by overlap.

Annual totals were obtained by aggregating hour values from monthly files by year and category. The data-quality report checked filename–month correspondence, coordinate ranges, negative values, expected relationships between hour variables, and missing core fields. Agreement between monthly and independently aggregated annual totals, apart from very small floating-point differences, was used as a check on aggregation consistency. These checks do not validate model accuracy, the provider's classifications, or unobserved activity.

3. Results

3.1. Annual trends by flag category

Across the 150 MPA analytical units, CHN apparent fishing hours increased by 6.16% between 2022 and 2024, while UNKNOWN-CHN increased by 319.62%. The total net increase from 2023 to 2024 was 4.230.13 hours, approximately 83.7% of which was associated with UNKNOWN-CHN. Changes were spatially concentrated, and the increase in total apparent fishing hours occurred alongside a decline in AIS presence hours. The findings illustrate the value of spatial AIS-based monitoring with overlap correction, while underscoring the need to account for data coverage, flag assignment, grid resolution, data-processing versions, and MPA-specific rules.

UNKNOWN-CHN apparent fishing hours increased from 4.631.42 in 2022 to 15.893.27 in 2023 and 19.434.50 in 2024, a 319.62% increase relative to 2022. AIS presence hours for this category were 15.335.02 in 2022, 35.308.56 in 2023, and 33.135.20 in 2024. UNKNOWN-CHN's share of the combined CHN and UNKNOWN-CHN apparent fishing hours rose from 3.15% in 2022 to 11.39% in 2024. Percentage increases should be considered alongside the relatively small 2022 baseline.

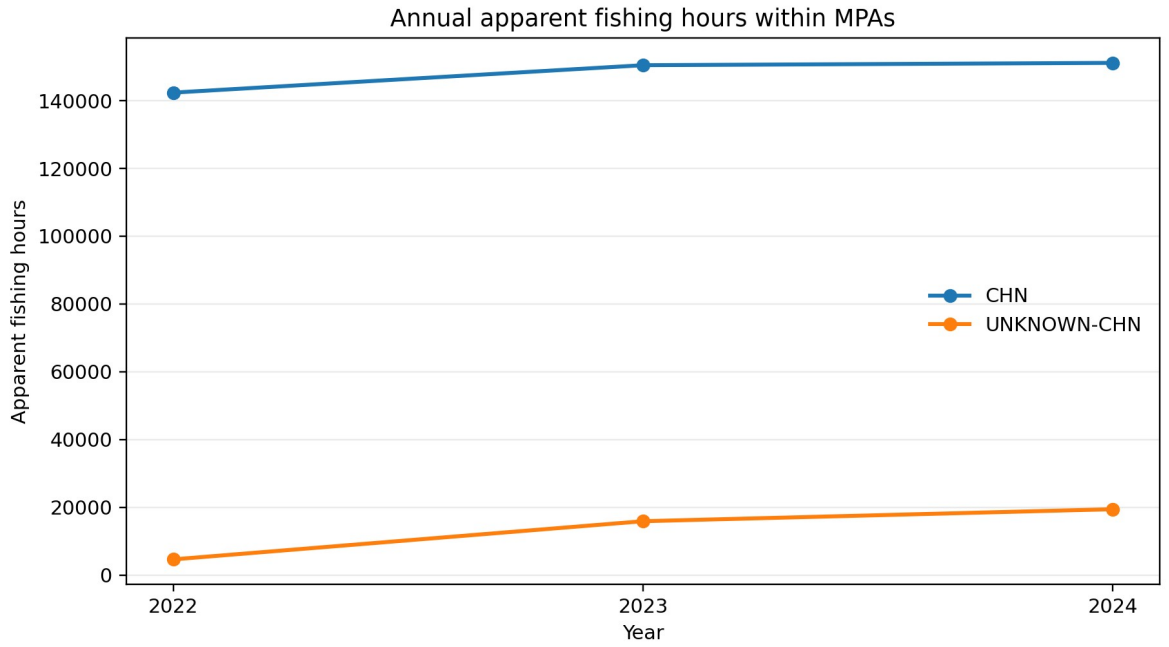


Figure 1. Annual apparent fishing hours for the CHN and UNKNOWN-CHN categories across MPA analytical units.

Year	Flag	Fishing hours	AIS presence hours	Fishing/AIS (%)	Change in fishing hours relative to 2022 (%)
2022	CHN	142.376.55	292.856.94	48.62	0.00
2023	CHN	150.462.06	358.639.69	41.95	5.68
2024	CHN	151.150.97	354.717.42	42.61	6.16
2022	UNKNOWN-CHN	4.631.42	15.335.02	30.20	0.00
2023	UNKNOWN-CHN	15.893.27	35.308.56	45.01	243.16
2024	UNKNOWN-CHN	19.434.50	33.135.20	58.65	319.62

3.2. Monthly and quarterly patterns

The highest monthly CHN totals occurred in September in 2022 and 2023 (23.213.42 and 19.240.36 hours, respectively) and in August in 2024 (17.055.05 hours). UNKNOWN-CHN monthly peaks occurred in September 2022 (1.320.93 hours), April 2023 (3.222.58 hours), and October 2024 (4.132.10 hours). The differing peak months indicate that the two categories did not follow the same seasonal pattern.

The third quarter was the highest quarter for CHN in all three years, although its share of annual activity declined from 43.20% in 2022 to 34.54% in 2023 and 33.32% in 2024. By contrast, the first-quarter share rose from 5.55% in 2022 to 17.36% in 2024. For UNKNOWN-CHN, the fourth quarter had the largest share in each year. In 2024, the third and fourth quarters together accounted for 77.59% of annual UNKNOWN-CHN apparent fishing hours. August–October 2024 alone represented 54.97% of the category's annual total.

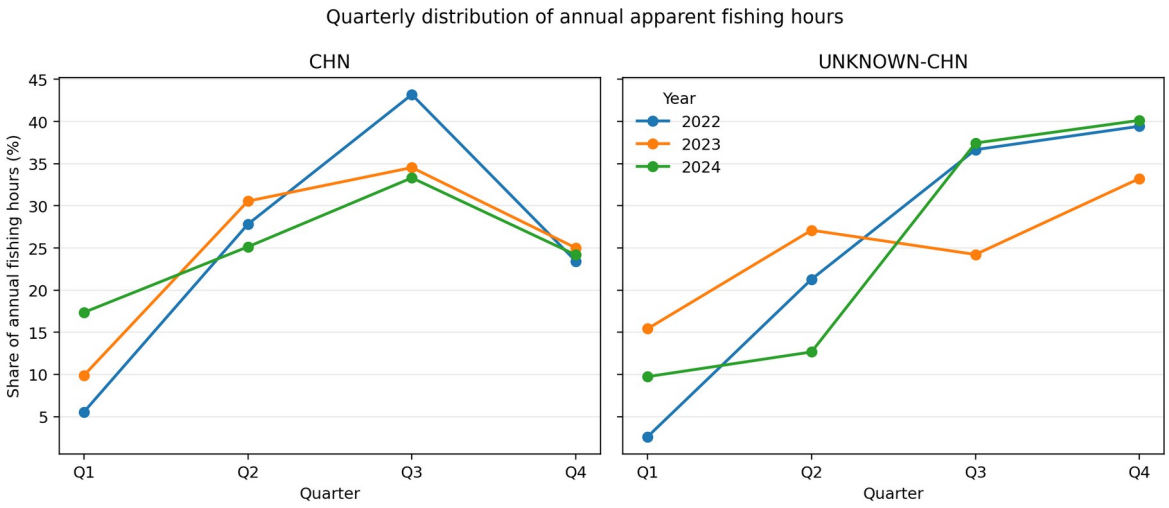


Figure 2. Quarterly distribution of annual apparent fishing hours by flag category.

3.3. Spatial concentration and changes across MPA units

Activity was concentrated in a limited number of MPA analytical units. For CHN, the five highest-activity units accounted for 94.05% of the total in 2022, 92.53% in 2023, and 86.56% in 2024. Thus, concentration among the top five units remained high but declined over the study period. UNKNOWN-CHN was distributed even more unevenly: Guangdong Nanpeng alone accounted for 85.28% of 2024 UNKNOWN-CHN apparent fishing hours.

Across all 150 units, combined CHN and UNKNOWN-CHN fishing hours rose from 166.355.34 in 2023 to 170.585.47 in 2024 (+4.230.13 hours; 2.54%). CHN contributed +688.91 hours (0.46%), while UNKNOWN-CHN contributed +3.541.22 hours (22.28%). Thus, approximately 83.7% of the net increase was associated with UNKNOWN-CHN. This decomposition is based on GFW flag categories and does not identify vessel ownership or nationality.

Among the 49 units with positive fishing hours in 2024, 34 increased and 15 decreased. Positive changes summed to 19.399.65 hours, while the absolute total of negative changes was 14.513.95 hours. The three largest increases occurred in Grand Sud du Gabon (+6.040.24 hours), Guangdong Nanpeng (+5.025.06 hours), and Phoenix Islands (+2.851.91 hours); together, these represented 71.74% of all positive changes. Marae Moana alone accounted for 69.49% of the absolute negative changes (−10.085.37 hours). The +4.885.69-hour net increase in the 2024-positive subset exceeded the increase across all units because 101 units with zero hours in 2024 had recorded 655.56 hours in 2023.

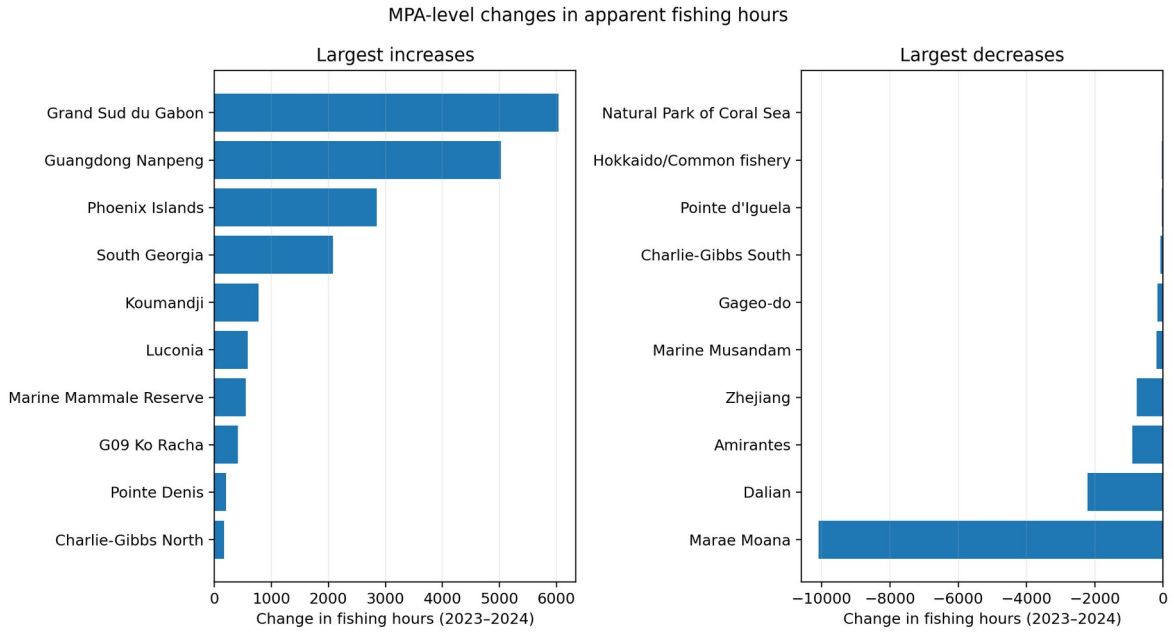


Figure 3. Largest positive and negative changes among units with positive fishing hours in 2024.

Category	2023 fishing hours	2024 fishing hours	Net change (hours)
CHN	150.462.06	151.150.97	+688.91
UNKNOWN-CHN	15.893.27	19.434.50	+3.541.22
Toplam	166.355.34	170.585.47	+4.230.13

Rank	Largest increase	Change (hours)	Largest decrease	Change (hours)
1	Grand Sud du Gabon	+6.040.24	Marae Moana	−10.085.37
2	Guangdong Nanpeng	+5.025.06	Dalian	−2.212.44
3	Phoenix Islands	+2.851.91	Amirantes	−902.60
4	South Georgia	+2.074.34	Zhejiang	−764.33
5	Koumandji	+775.41	Marine Musandam	−186.14
6	Luconia	+584.89	Gageo-do	−157.08
7	Marine Mammale Reserve	+546.18	Charlie-Gibbs South	−77.23

3.4. Relationship between fishing hours and AIS presence hours

Across all 150 units, total AIS presence hours fell from 393.948.25 in 2023 to 387.852.62 in 2024 (−6.095.63 hours; −1.55%). Apparent fishing hours increased over the same period. The aggregate fishing/AIS ratio rose from 42.23% to 43.98%. Of the 49 units with positive fishing hours in 2024, 27 showed increases in both fishing and AIS hours, while 11 showed decreases in both. In seven units fishing hours increased as AIS hours declined; in four, the reverse occurred. Unit-level changes were strongly positively correlated (Pearson r = 0.8501; n = 49).

The correlation indicates that changes in AIS presence and apparent fishing can move together across units, but it does not require the two aggregate indicators to move in the same direction. Among the 42 units for which the ratio could be calculated, the fishing/AIS ratio rose in 24 and fell in 18. Very large percentage-point changes sometimes resulted from very small 2023 baselines. Ratios should therefore be considered alongside absolute hours.

3.5. Local examples

All three analyzed 0.1-degree cells associated with Guangdong Nanpeng showed both CHN and UNKNOWN-CHN activity in all three years. In this selected three-cell area, the increase was associated with intensification of apparent activity in the same cells rather than spatial expansion into new cells. Total fishing hours rose from 10.674.26 in 2022 to 23.048.80 in 2023 and 28.073.86 in 2024. UNKNOWN-CHN's share of the total rose from 21.7% in 2022 to 59.0% in 2024. This local example is consistent with Guangdong Nanpeng's large contribution to the overall UNKNOWN-CHN pattern.

The Phoenix Islands example demonstrates the importance of overlap correction. Two highly similar MPA records shared a large proportion of their grid observations. After deduplication, fishing hours in the combined PIPA area rose from 16.906.17 in 2023 to 19.758.08 in 2024 (+2.851.91 hours; +16.87%). Of the cells with positive activity in 2024, 620 had zero apparent fishing hours in 2023 and positive values in 2024; in 291 cells, values were positive in 2023 and zero in 2024. These are changes in observations at the grid-cell scale; they do not demonstrate that fishing began or ceased in the real world.

When interpreting the scale of these findings, total hours should not be equated with vessel counts. The totals aggregate periods assigned by the model to particular cells and time intervals; prolonged presence by the same vessel can increase the total. GFW also describes the 2024 data as provisional because vessel classifications may change as data from later periods become available (Global Fishing Watch, 2025a). Comparisons involving the final year should therefore be checked by keeping the data-product version fixed and recalculating them when updated files become available.

4. Discussion

4.1. Different patterns across flag categories

The findings show different trajectories in apparent fishing activity for the two GFW categories across the MPA analytical units. CHN showed a modest increase over the three-year period, whereas UNKNOWN-CHN increased much more rapidly from a smaller baseline. The contribution of UNKNOWN-CHN to the net increase from 2023 to 2024 is notable, but the category must be interpreted according to GFW's assignment logic. It does not directly identify Chinese-flagged vessels, Chinese beneficial ownership, or a specific fleet.

4.2. Spatial concentration and aggregation effects

The marked spatial concentration matters for monitoring and management. A small number of analytical units can have a substantial effect on category totals, meaning that changes in one or two locations may materially alter the overall total. Guangdong Nanpeng's large contribution to UNKNOWN-CHN and the marked decline in Marae Moana show how overall or regional averages can conceal local differences. Monitoring priorities should therefore consider unit-level patterns and local management conditions, as well as aggregate values.

4.3. Reading fishing and AIS presence indicators together

The increase in fishing hours alongside a decline in AIS presence hours from 2023 to 2024 warrants careful interpretation. Although the unit-level correlation is strong, aggregate values moved in different directions because units differed in baseline size and direction of change. The fishing/AIS ratio can help describe the share of AIS presence time classified by the model as fishing. It cannot be used as evidence of fishing efficiency, catch rate, or more intensive resource extraction.

4.4. Data-production changes and uncertainty

Temporal comparisons may also be affected by changes in data production. GFW reports improved AIS reception in some high-density areas of China and Southeast Asia following the introduction of dynamic receivers around 2022. Version 3 of the data-processing pipeline also included vessel-classification changes relative to the previous version, affecting apparent fishing-hour estimates. These factors can create differences unrelated to fishing behavior. Future studies should retain the full archive name, download date, README files, known-issues documentation, and schema, and should avoid mixing data-processing versions in a time series.

Spatial assignment is limited by the 0.1-degree grid resolution. Under the project's cell-center point-in-polygon approach, a cell is associated with an MPA when its center lies inside the polygon. The entire cell may not lie within the MPA boundary, and activity in other parts of the cell cannot be assigned definitively to the protected area. The physical area of 0.1-degree cells also varies with latitude. Area-weighted allocation or higher-resolution data could improve boundary precision, but would introduce additional assumptions and should be assessed through sensitivity analysis.

4.5. Implications for protected-area management

Finally, the presence of apparent activity within an MPA polygon does not by itself constitute a legal violation. MPAs differ in zoning, permitted uses, effective dates, and enforcement regimes. A "Not Reported" no-take value in a database indicates missing or unreported information; it is not evidence that fishing is either prohibited or permitted. Assessing compliance requires separate examination of the relevant legal instruments, spatial zoning, and temporal validity.

For management, these results should primarily be used to identify locations that warrant further examination. In units that dominate a category total, such as Guangdong Nanpeng, flag assignment, signal coverage, and local fishing context could be checked against additional sources. In locations with marked declines, such as Marae Moana, changes in AIS reception and the cells covered by the area boundaries could be reviewed. This approach can guide field review and data validation without turning hour totals directly into enforcement indicators.

5. Limitations

This study is descriptive and relies on AIS-based model estimates. It does not directly observe catch volume, actual gear deployment, vessel intent, or legal status. AIS participation and signal reception are incomplete and spatially uneven. UNKNOWN-CHN is not a verified nationality category. The three-year period is relatively short and coincides with changes in AIS reception and the GFW data-processing pipeline. The 0.1-degree grid limits boundary precision. Analytical units were constructed to address overlapping MPA records and may not in every case represent a single legal management jurisdiction. The Pearson correlation is descriptive for the 49 units with positive fishing hours in 2024 and should not be generalized to all MPAs without further analysis.

In addition, the scope and characteristics of MPA records depend on the WDPCA version used; names, boundaries, year of status, and management attributes may be updated. Because the category-level analysis does not track vessel identities, the same vessel's activity across multiple months or cells may appear in separate units. The selected three-cell Guangdong example and the Phoenix Islands analysis are illustrative local cases; they are not random samples or a subset representative of all MPAs. The Pearson correlation may also be influenced by outliers and was calculated only for the 49 observed units. The coefficient is therefore reported as evidence of strong co-variation, not as a significance test or a basis for generalization.

5.1. Directions for future research

Future studies could test the observed patterns in several ways. First, the time series could be rebuilt using updated versions, with known GFW version changes reported as separate sensitivity scenarios. Second, the cell-center assignment approach could be compared with cell–polygon intersection-area weighting or alternative resolutions to quantify boundary effects. Third, vessel-level AIS data, independent VMS records, regional observations, and local regulations could be linked to reduce uncertainty in categories and models. Finally, longer time series for the same MPAs and suitable comparison areas would provide a stronger design for distinguishing seasonality from possible management effects.

The findings of this study can help set monitoring priorities and guide follow-up analyses. However, assessing conservation performance requires independent measures such as ecological condition, pressure indicators, management practices, and legal zoning. The apparent fishing hours reported here are not substitutes for those measures.

6. Conclusion

Across the 150 MPA analytical units, CHN apparent fishing hours increased by 6.16% between 2022 and 2024, while UNKNOWN-CHN increased by 319.62%. The total net increase from 2023 to 2024 was 4.230.13 hours, approximately 83.7% of which was associated with UNKNOWN-CHN. Changes were spatially concentrated, and the increase in total apparent fishing hours occurred alongside a decline in AIS presence hours. The findings illustrate the value of spatial AIS-based monitoring with overlap correction, while underscoring the need to account for data coverage, flag assignment, grid resolution, data-processing versions, and MPA-specific rules.

Author contribution statement

Sude Nur Kılıç: data curation, computational data processing, spatial analysis, formal analysis, visualization, and preparation of the initial manuscript draft. The final contribution statement should be aligned with the target journal’s taxonomy and updated to reflect any other contributors.

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